

2020 AMC 10A Problem 21 - Solution

Review the full statement and step-by-step solution for 2020 AMC 10A Problem 21. Great practice for AMC 10, AMC 12, AIME, and other math contests

2020 AMC 10A Problem 21

Problem:

There exists a unique strictly increasing sequence of nonnegative integers $a_1 < a_2 < \dots < a_k$ such that

$$\frac{2^{289} + 1}{2^{17} + 1} = 2^{a_1} + 2^{a_2} + \dots + 2^{a_k}$$

What is k ?

Answer Choices:

- A. 117
- B. 136
- C. 137
- D. 273
- E. 306

Solution:

First, substitute 2^{17} with x . Then, the given equation becomes

$$\frac{x^{17} + 1}{x + 1} = x^{16} - x^{15} + x^{14} - \dots - x^1 + x^0$$

by sum of powers factorization. Now consider only $x^{16} - x^{15}$. This equals

$$x^{15}(x - 1) = x^{15} \cdot (2^{17} - 1)$$

Note that $2^{17} - 1$ equals $2^{16} + 2^{15} + \dots + 1$, by difference of powers factorization (or by considering the expansion of $2^{17} = 2^{16} + 2^{15} + \dots + 2 + 2$). Thus, we can see that $x^{16} - x^{15}$ forms the sum of 17 different powers of 2. Applying the same method to each of

$$x^{14} - x^{13}, x^{12} - x^{11}, \dots, x^2 - x^1$$

we can see that each of the pairs forms the sum of 17 different powers of 2. This gives us $17 \cdot 8 = 136$. But we must count also the x^0 term. Thus, Our answer is $136 + 1 = \boxed{(C) 137}$.

OR

Multiply both sides by $2^{17} + 1$ to get

$$2^{289} + 1 = 2^{a_1} + 2^{a_2} + \dots + 2^{a_k} + 2^{a_1+17} + 2^{a_2+17} + \dots + 2^{a_k+17}$$

Notice that $a_1 = 0$, since there is a 1 on the LHS. However, now we have an extra term of 2^{18} on the right from 2^{a_1+17} . To cancel it, we let $a_2 = 18$. The two 2^{18} 's now combine into a term of 2^{19} , so we let $a_3 = 19$. And so on, until we get to $a_{18} = 34$. Now everything we don't want telescopes into 2^{35} .

We already have that term since we let

$$a_2 = 18 \implies a_2 + 17 = 35$$

Everything from now on will automatically telescope to 2^{52} . So we let a_{19} be 52.

As you can see, we will have to add $17a'_n$'s at a time, then "wait" for the sum to automatically telescope for the next 17 numbers, etc, until we get to 2^{289} . We only need to add a'_n 's between odd multiples of 17 and even multiples. The largest even multiple of 17 below 289 is $17 \cdot 16$, so we will have to add a total of $17 \cdot 8a'_n$'s. However, we must not forget we let $a_1 = 0$ at the beginning, so our answer is

$$17 \cdot 8 + 1 = \boxed{(C) 137}$$

The problems and solutions on this page are the property of the MAA's [American Mathematics Competitions](#) ↗

Powered by [Wiki.js](#)